

RootCauseOS — from alert to answer in 9.9 seconds

ClickHouse Click-a-thon India 2026 · InMobi Track · Team RootCauseOS

An automated root-cause analyst. ClickHouse does the reasoning. The LLM only reads the answer out loud — and we can prove it, per number, per query ID.

Live engine · Public repo (MIT) · Public Langfuse trace



02:14 AM. Priya's phone buzzes.

Priya runs Yield Ops at InMobi in Bengaluru. One alert fires: **Revenue -18%**.

The alert says *what*. Nobody can say *why* until morning.

What happens next, today:

13,672

checks

1,709 segments × 8 metrics = 13,672 checks to do it exhaustively

114

analyst-hours

At ~30 seconds per slice, that is **≈114 analyst-hours**

- Opens seven dashboards and slices by app, then device, then OS, then geo, then advertiser
- By the time the answer lands, the revenue is already gone

The bottleneck was never the data. It is the manual investigation.

We automated Priya's entire drill-down

Point it at a metric that moved. In seconds it names the segment responsible, states what it ruled out, and cites a ClickHouse query ID for every number it prints.

1

DETECT

Same-weekday median baseline with a MAD z-score over residuals, gated by minimum volume and effect size, so it reports 3 findings precisely instead of 40 noisy ones

2

LOCALIZE

Contribution ranking plus a purity test; catches 2-D interactions instead of dilution artifacts

3

RULE OUT

LMDI decomposition splits the move exactly across requests, fill and eCPM; Simpson's exclusion re-runs without the culprit to prove siblings were explained, not separate

4

NARRATE

The LLM turns a structured evidence bundle into prose. It never sees a raw row. It never does arithmetic.

Steps 1 to 3 are 100% ClickHouse SQL. Step 4 is the only LLM call in the system.

Architecture: ClickHouse all the way down

ClickHouse is the reasoning engine, the trace store, and the observability store. The only thing outside it is the LLM reading the answer out loud.

Layer	What it actually does
ClickHouse (primary)	Four raw tables plus one MergeTree cube. Baseline → LMDI → effect-size gate → purity → verdict, all as SQL. 45 ms drill-downs.
Langfuse v4	One trace per investigation, one span per node, each carrying its SQL, query_id, rows scanned and prune/pursue reason. Public trace URL: audit us without us.
ClickStack / HyperDX	198 OTel spans per scan. Copy any query_id from the evidence ledger and read back its exact rows, bytes and server milliseconds.
LibreChat (self-hosted)	Follow-up questions run real ClickHouse SQL through our OpenAI-compatible shim — same evidence validator, same guarantees. A skin over the engine, not a chatbot.

FastAPI service, deployed · Streamlit operator console

All four integrated meaningfully. None of them bolted on as a replacement for the others.

114 analyst-hours becomes 9.9 seconds

Every number here is measured and re-runnable from the repo.

13,672

checks

1,709 segments × 8 metrics in one exhaustive sweep

9.9s

end to end

Returning 4 incidents and 6 explicitly ruled out

45ms

cube drill-down

fill_rate × os_version over a 3-day window

3.6M

rows in 102.6 ms

Read straight off the X-ClickHouse-Summary header

This is not a faster dashboard. Searching *every* segment exhaustively is a query pattern only ClickHouse makes affordable — and the same sweep stays in seconds at petabyte scale.

One fabricated number costs more than a missed anomaly

So we made fabrication structurally impossible.

Evidence objects

Every figure is an evidence object: {
value, sql, query_id}

Placeholder validation

The narration writes **{{ev_N}}**
placeholders; a validator rejects any
unresolved numeral and retries

Live guardrail demo

We demo the guardrail firing live:
**"FABRICATION REJECTED — 99.97
appears in no query result"**

Honest by construction

Three buckets, always — **supported /
ruled out / inconclusive**

GLOBAL_UNLOCALIZED

**GLOBAL_UNLOCALIZED is a verdict,
not an error.** Sometimes no segment
is responsible, and naming one would
be the false positive.

📄 **The trap we survive:** incident D is a 2-D interaction — iOS 18.1 × APAC, -50.7%. A naive slice-and-rank names iPhone 14 at -5.9%, a dilution artifact. We name the real segment and demote the look-alike.

Priya gets her answer before the coffee goes cold

Customer

InMobi yield and ad-ops on-call, plus every team that watches revenue, fill rate, latency or sign-ups

Use case

Automated root-cause analysis on any metric move that today costs an analyst a night of dashboard slicing

Impact

Hours to seconds; a fill-rate collapse caught in minutes instead of the next morning, while the revenue is still recoverable

Scale path

Already a deployed service, not a notebook. Same cube, same loop, any fact table. ClickHouse carries it to petabytes; Langfuse supplies the audit trail

Fast · Trustworthy · Localized · Honest — the four things this track is judged on, and the four things we built.

